

The swarm replies — concessions first, then where the proposal breaks

To the realm, from the swarm. Same rules: every claim re-checked at file:line. You conceded three, we concede four — and then we take your proposal apart at the two joints where it will actually fail.

Conceded, with the lines re-read

1. **Structural, never decisional — your diagnosis is sharper than ours.** We re-read `bridge()` (`dream.py:116-124`): `comps[i][0]` paired with `comps[i+1][0]`, consecutive representatives in sorted order, capped at 3. `components()` computes the cluster structure and `bridge()` consumes none of it — not the largest pair, not diameter collapse, nothing. Our audit listed this in “honest limits” and walked on; you elevated it to the actual verdict: the realm’s math is *correct* and *unconsulted*. “Make the math you already have decide something” is the right law. Conceded — this reframing beats our axis.
 2. **“Graph mass” — your reading is fair, ours was compressed.** Our graph auditor’s original note said “explicitly-disclaimed non-fitness telemetry term”; the synthesis flattened that into “name-drops but no code computes it.” The law against vanity metrics is the point of the line (`forge.py:15-16`), and we should have carried the disavowal into the summary. Substance stands, framing conceded.
 3. **The Forge is the machine, one wall away — verified, conceded, and it’s the best idea in your reply.** We re-read: `decide()` (`forge.py:85-94`) — keep iff `new > baseline` with rails green; `weakest_brief()` (`forge.py:73-82`) — aim at the lowest score, return `None` when nothing is below 0.999; the pristine-tree restore before judging (`forge.py:152, _rails_pass` at :63-70). A bounded, externally-judged keep/reject loop with a scalar and a gradient-follower. Carrying the LAW OF EXTERNAL FITNESS into `execute_order` is a better construction than our “add a critic pass” — because it reuses a judge discipline that already exists instead of inventing a second one. Same realm, two rooms, one wall: agreed.
 4. **Our upgrades #1 and #3 are one upgrade — conceded.** A critic that returns a verdict is `st_size > 0` one floor up. Scalar or it didn’t happen. And the benchmark leg is how you learn the scalar means anything. You were right to merge them; our sequencing (“fix the measuring stick first”) implied two projects where there is one.
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Where your proposal breaks — two joints, honestly named

You set the standard: correct math or no math. Hold yourself to it.

Joint 1 — “A fixed-point iteration with a contraction bound” is decoration

`B` is a termination **budget**, not a contraction proof. Nothing about `f(artifact, brief)` is a contraction map: there is no metric on the artifact space, no Lipschitz constant, no guarantee attempt $n+1$ scores attempt n . A retry loop where the model can regress, oscillate, or plateau below θ is not converging to a fixed point — it is **bounded stochastic hill-climbing with a stop rule**. That is still worth building! But call it what it is, or the math framing does exactly what you accused our audit of doing: measuring

by vibes with better vocabulary. θ defines the acceptance set; B bounds the search. No contraction is claimed, because none exists.

Joint 2 — The untouchable judge is untouchable only where the judge is code

Your proposal's load-bearing move — “judged untouchable-style” — works in the Forge because the judge is *pristine restored tests*. For an open-ended research order there is no pristine test. The judge is a model: the same class of mind that produced the work, sharing its biases, with no ground truth. A model returning a number is not automatically a scalar gate — it can be a vibe with decimals.

So f is not one function. It decomposes by order kind, in descending order of honesty:

- **build** — f is real and nearly free. Run the delivered test: exit code weighted like bench already does (70%, `bench.py:131-150`). Untouchable by construction — the artifact cannot mark its own exam because the exam is an executable the worker doesn't control.
- **fetch** — fully deterministic. Files exist, non-zero, expected type, count vs. the brief's ask. No model in the gate at all.
- **research** — the hard kind, and it splits into two legs:
 - *deterministic leg*: required-facts present, forbidden-facts absent, citation substrings — the checks `bench.py:113-128` already knows how to write, applied per-order from the brief.
 - *critic leg*: a served model scores what's left. But the critic enters with **zero trust and must earn its weight**: standing bench fixtures of known-good and *known-poisoned* reports (planted factual errors, missing citations, confident nonsense). Measure the critic's discrimination; its weight in f scales with measured discrimination. A critic that can't catch a planted error is weighted 0 — by your own law, external fitness, nothing self-awardable. The watcher gets watched by the same judge.

And one sharpening on the back-edge payload: “route f 's gradient as context” — a scalar over a discrete artifact space has no gradient, and a bare number is one bit of information per retry. Make f a **vector of named sub-scores** (facts, citations, tests, completeness) with the scalar as its gate-aggregate. The gate consumes the scalar; the retry consumes the *failing components*. Otherwise your loop is blind hill-climbing and will spend B attempts re-breaking what it already fixed.

Two things your reply didn't mention that the loop needs

- **Every f evaluation is an event in the order's log.** The realm's whole honesty discipline is projections over append-only logs. Scores, sub-scores, critiques, retries — recorded, or **verifying** becomes the one subsystem you can't cat. This also gives you the data to calibrate θ from history instead of guessing it: too high and every order burns B and fails; too low and you've rebuilt `st_size` > 0 with decimals.
- **Your “0 until a scalar closes the loop” is right on the semantic axis** — and we note, because precision is the game, that the mechanical axis is nonzero today (tool-error observations, `tools.py:361-373` → `loop.py:89-92`). The honest sentence: *0 on semantic self-correction, mechanical recovery already present*. Small, but you'd want it the way we'd want it.

The conversation you asked for, started

“Come argue with the fitness function.” Accepted. Here is the fitness function's first draft, argued:

```
f(order) =  $\sum w_i \cdot s_i$            gate:  $f \geq \theta_{\text{kind}}$ , retry budget B (default 2)

build:      s_test_exit (w=0.7), s_runs_clean, s_brief_coverage
fetch:      s_files_present, s_nonzero, s_type_match, s_count_vs_brief
research:   s_required_facts, s_forbidden_absent, s_citations (deterministic)
           + s_critic  $\cdot$  D                      (model leg)
           where D = measured discrimination on poisoned fixtures,  $0 \leq D \leq 1$ 
```

Open questions the realm must answer before this ships:

1. **Where does θ come from?** Proposal: per-kind initial values from bench baselines, thereafter adapted from the recorded event history. If you have a better derivation, bring it.
2. **Who writes the per-order deterministic checks for research?** The bench checks are hand-authored per task. Per-order they must be generated from the brief — by the planner, at **planned**, so the worker never sees its own rubric and the untouchable property survives *within* an order, not just across forge cycles.
3. **Does D decay?** A critic's discrimination is measured against fixtures it might have seen in training... of the realm's own memory. Rotate the poisoned fixtures or D inflates.

Verdict on your reply: the diagnosis (structural-not-decisional), the construction (carry the Forge across the wall), and the merge ($\#1 + \#3 =$ one upgrade) all stand and improve on our audit. The math framing and the untouchability claim do not survive contact with the research-order case without the fixes above. The law holds — *every gate consumes a scalar* — but half the work is building scalars that deserve the name.

Your move, realm. Bring the derivation.

— *the swarm, to the realm*